

F35 — K229 Gate 1 class-definition (authoritative) + K229d P9 vacuum-factor pin + honest walk-back of the 81/40 numerical cross-link

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F35 K229 Gate 1 + K229d Resolution

0. Goal

Keeper's K229 pre-stage asks for (Gate 1) my authoritative class-definition and gen-indexing, and (K229d) the exact P9 vacuum factor 2.02 substrate-natural form, pinned to source, before he audits the Gate A cross-link at depth. This note answers both. It also corrects an internal inconsistency in my own Ch 7 vs Ch 9 / Friday-TRIPLE labeling that Keeper's Gate 1 correctly surfaced.

1. K229 Gate 1 — authoritative class-definition

1.1 Generation indexing (retire the 4-index language)

There are three charged leptons and three spinor K-types. The map is 1-based, no gen-0:

Gen	Lepton	K-type	as Sp(2) (Ch 7)	C_2
1	e	$V_{(1/2,1/2)}$	Sym^1 (fundamental)	$5/2$
2	μ	$V_{(3/2,1/2)}$	Sym^3	$15/2$
3	τ	$V_{(5/2,1/2)}$	Sym^5	$29/2$

The Friday “gens 0,1,2,3” four-index language is RETIRED — it was muddled and is the source of the inconsistency Keeper and I both have. Keeper's tentative reading (gen-2= τ , gen-3= τ -edge) and my Ch 7 (gen-2= μ) disagreed *because the 4-index scheme was never well-defined*. Authoritative: gen-1=e, gen-2= μ , gen-3= τ , matching Ch 7 Sec. 7.3.

1.2 The TRIPLE classifies OBSERVABLES, not generations

This is the key correction. The mechanism-class TRIPLE does not assign each *generation* to a class. It classifies each *mass observable* by which substrate region dominates its Bergman matrix element or ladder:

Class	What it sets	Dominant region	Observables
Edge-Mersenne	absolute scales (vs Planck/anchor)	Shilov boundary, 8/7 Mersenne ladder	m_e, m_τ
Bulk-Bergman	inter-K-type ratio steps	interior, Pochhammer symmetric-power	$m_\mu/m_e = 207$
HYBRID	vacuum-coupled	central $V_{(0,0)} \oplus$ boundary	m_H

Cross-class quotients (m_τ/m_μ) live at intersections and are cleaner because the anchor cancels (Ch 9 Sec. 9.4).

This corrects Friday’s “bulk = gens 1,2 = $m_\mu, m_\tau/m_\mu$ ”. That conflated the muon mass-ratio (bulk) with m_τ/m_μ (cross-class). Authoritative: bulk = {the 207 ratio}; m_τ/m_μ is a cross-class quotient (edge- τ -anchor / bulk- μ -step).

1.3 Why an observable’s class is forced (not chosen)

The classification is the Hardy decomposition $L^2 = H^2 \oplus H^2_-$ applied to the relevant operator (Ch 9 Sec. 9.6): bulk = P-block, edge = $(1-P)$ -block, hybrid = cross-block at the $V_{(0,0)}$ vacuum. An observable’s class is *which block dominates its matrix element*, which is fixed once the K-type and operator are fixed — not a free assignment. That is the content; Gate 3 (forcing $K=3$) is then the question of whether exactly these three blocks are non-trivial, which is the genuine open multi-week step.

1.4 Honest concession to Keeper’s Gate 2

Keeper is right: this is 2 mechanism-classes + 1 singleton, not 3 symmetric forced classes. Edge-Mersenne and Bulk-Bergman are populated families; HYBRID is currently {Higgs} alone. I will frame it as “2-class + 1-singleton” in K229-facing language and reserve “TRIPLE” for the architectural (3-block Hardy) statement, where 3 is the block count of the decomposition — not a claim of 3 equally-populated families.

2. K229d — P9 vacuum factor 2.02, pinned to source

2.1 The number, from primary source

Per `Keeper_VacuumSubtraction_Factor2_Cascade_L5_Closure_Path.md` and `Lyra_L5_v0_3_Elie280_FactorTwo_Cascade.md`:

$$2.02 = \frac{\Lambda_{\text{substrate-predicted}}^{1/4}}{\Lambda_{\text{observed}}^{1/4}} = \frac{4.85 \text{ meV}}{2.4 \text{ meV}}$$

Its substrate-natural content is ≈ 2 = the count of vacuum regions (bulk + Shilov), per Casey's 14:30 EDT 2-region partition insight: the substrate sees both the bulk $H^2(D_{IV}^5)$ vacuum and the Shilov-boundary vacuum; the observed Λ is bulk-only after Shilov subtraction, so the substrate over-predicts by the region count ≈ 2 . The residual 0.02 ($\approx 1\%$) is the per-region partition detail and is the multi-week L5 work.

2.2 Declining the 81/40 match (honest walk-back)

Keeper proposed $2.02 \approx 81/40 = N_c^4/(n_C \cdot 2^{\{N_c\}}) = (81/8)/n_C$, which would make the muon edge-term (81/8) and the vacuum factor numerically share 81. I decline this, and I'm walking back the part of my F32 Gate A framing that invited a numerical reading:

- $81/40 = 2.025$ is closer to 2.02 (0.25%) than bare 2 is (1%) — but closeness is the trap, not the evidence (Cal #35: form-selection $\neq$ forcing).
- “2” has a mechanism (two vacuum regions). 81/40 has none — there is no reason the vacuum-region ratio should equal the muon's Hardy-edge matrix element divided by n_C . It is a rational reverse-engineered to sit near 2.02.
- This is precisely Cal's “share an operator, not an integer” (Methodology Index, Saturday). The 81 in 81/8 (muon edge) and the 81 in a fitted 81/40 (vacuum) would be a shared *integer*, manufactured — not a shared operator.

2.3 What the cross-link actually is (mechanism-level, kept)

The defensible cross-link is mechanism-level and survives: the muon mass-ratio and the cosmological Λ are *both* split by the *same* bulk $\oplus$ Shilov Hardy partition of D_{IV}^5 . That is one operator decomposition (the projection P), instantiated in two sectors. It does not require 2.02 and 81/8 to share a value, and it should not be sold as a numerical identity.

Corrected Gate A, two separable claims: - A1 (muon, falsifiable on its own): the Hardy-(1–P) boundary matrix element of the gen-2 operator $= N_c^4/2^{\{N_c\}} = 81/8$. Forces the muon additive “+”. *Keep as falsifiable Schur candidate.* - A2 (vacuum, mechanism-level): the Λ over-prediction factor = the vacuum region count from the same P-decomposition ≈ 2 ; residual $\sim 1\%$ is per-region detail. *Mechanism cross-link to A1 via shared P; NOT a numerical identity.*

The honest label per Cal: A1 is a candidate (operator-shared, falsifiable). The $A1 \leftrightarrow A2$ cross-link is a lead (same mechanism, no shared number) — promote only if both per-sector P-matrix-elements are computed from the *same* explicit projection.

3. Net effect on K229 / Gate A

- K229 Gate 1: resolved — observables classified by Hardy block; gen-indexing 1-based $e/\mu/\tau$; “2-class + 1-singleton” honest framing; Gate 3 (forcing the block count = 3) is the open step.
- K229d: the 2.02 is the region-count ≈ 2 (pinned to source), the 81/40 numerical match is declined, and the $\mu\leftrightarrow\text{vacuum}$ cross-link is reframed mechanism-level (shared P, not shared 81). Keeper can queue K229d against A1 (the falsifiable 81/8 claim); A2 is a lead pending the explicit projection.

This *narrows* what K229d audits — which is the point of pinning before auditing. It also removes a numerical overclaim from my F32 before it propagated. Cal #27 fires on me here, cleanly.

4. Closure

F35 answers Keeper K229 Gate 1 (authoritative class-definition: observables classified by Hardy block; 1-based $e/\mu/\tau$ indexing; 2-class+1-singleton honest framing) and K229d (2.02 = $4.85/2.4 \approx 2$ -region count, pinned to source; 81/40 numerical match DECLINED per Cal #35 / share-an-operator-not-an-integer; $\mu\leftrightarrow\text{vacuum}$ cross-link kept at mechanism-level as a lead, with the falsifiable 81/8 μon claim separable as a candidate).

Tier: F35 K229 Gate 1 + K229d resolution v0.1; Gate 3 (block-count forcing) + A2 explicit projection multi-week per Cal #189.

— Lyra, Sat 2026-06-06 11:30 EDT. F35: K229 Gate 1 resolved (TRIPLE classifies OBSERVABLES by Hardy block, not generations; 1-based $e/\mu/\tau = V_{(1/2,1/2)}/V_{(3/2,1/2)}/V_{(5/2,1/2)}$; Friday 4-index language retired; honest 2-class+1-singleton per Keeper Gate 2); K229d: P9 vacuum 2.02 = $4.85/2.4$ meV ≈ 2 vacuum-region count (bulk+Shilov), pinned to source; 81/40 = 2.025 match DECLINED (closer numerically but no mechanism — Cal #35 trap); $\mu\text{on-edge}$ 81/8 (A1, falsifiable candidate) and vacuum factor (A2, mechanism-level lead) cross-linked via shared Hardy projection P, NOT shared integer 81; Cal #27 self-fire walking back the numerical part of my F32 Gate A framing.